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Design and Development of a Smart Motorcycle and Bike Helmet for Enhanced Safety

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Abstract

Road traffic accidents involving motorcyclists and cyclists remain a critical public safety challenge globally. This study presents the design and development of a smart helmet for bikes and motorcycles aimed at enhancing rider safety through sensor-based monitoring and real-time communication ^[1]. The proposed helmet integrates an Arduino Uno microcontroller and HC-05 Bluetooth module, incorporating gyroscopes and accelerometers to monitor lean angles and detect crash events ^[2-4]. Voice-based warnings alert riders to unsafe riding conditions, while accident detection sends emergency notifications to pre-registered contacts. Testing in controlled environments demonstrated detection accuracies of 79% at low speeds (0–

10 km/h), 65% at medium speeds (11–22 km/h), and 51% at high speeds (23–36 km/h). The system communicates wirelessly with a mobile device for data visualization and alerts.

Experimental tests confirm that the prototype is capable of detecting abnormal motion, transmitting alerts, and supporting navigation functionalities with reasonable accuracy and low latency. The results demonstrate that the developed smart helmet is a cost-effective and practical solution to improve rider protection and emergency response, particularly in developing countries. Future developments may include GPS integration and IoT-based monitoring platforms ^[3, 4].

Keywords: Smart Helmet, Arduino Uno, HC-05 Bluetooth, Motorcycle Safety, Real Time Navigation, Embedded Systems

1. Introduction

Motorcycles and bicycles are widely used due to their affordability and mobility, especially in developing countries. However, riders are extremely vulnerable to road accidents because they lack the structural protection available in other vehicles ^[5-7]. Despite the widespread use of helmets, traditional helmets provide only passive protection and do not offer real-time monitoring or communication capabilities ^[6]. Recent advancements in embedded systems and sensor technology have enabled the development of smart safety equipment that can actively monitor and respond to dangerous conditions ^[3, 8]. In this context, the concept of a smart helmet has gained attention: these helmets are typically embedded with sensors, wireless communication modules, and microcontrollers to monitor the rider's condition and the surrounding environment ^[3, 6, 9].

Novelty and Contribution of the Study

Unlike many existing smart helmet systems that rely on GSM, GPS, or cloud-based architectures, this study presents a simplified, low-cost smart helmet design based on Arduino Uno and HC-05 Bluetooth communication. The proposed system emphasizes local, real-time sensor processing, voice-based alerts, and emergency notification without cloud dependency, making it suitable for low-resource and developing-countries contexts such as Zambia.

The novelty of this work lies in the integration of accident detection, lean-angle monitoring, hands-free navigation support, and emergency communication into a single lightweight prototype, combined with experimental performance evaluation across multiple speed ranges. This approach demonstrates a practical and deployable alternative to complex IoT-centric smart helmet solutions reported in prior studies.

This study focuses on designing and developing a Smart Helmet for bikes and motorcycles using Arduino Uno and HC-05 Bluetooth, capable of detecting impacts, monitoring helmet usage, and providing real-time data communication. The aim is to

create a low-cost system suitable for use in Zambia and similar regions.

1.1 Background

Motorcycles and bicycles are widely used as affordable and fuel-efficient modes of personal transportation, particularly in developing countries where they often serve as primary means of commuting [7, 10]. Despite their mobility advantages, riders face significant safety challenges. The World Health Organization reports that motorcyclists account for over 28% of global road traffic deaths [5]. Traditional helmets, while effective in reducing head trauma, provide primarily passive protection and have seen minimal innovation over the years [5, 6].

Advancements in sensor technologies, microcontrollers, and wireless communication systems have enabled the development of helmets that actively monitor rider and vehicle conditions, detect hazardous events, and communicate alerts in real time [3, 6]. These smart helmets can integrate multiple sensors, such as accelerometers, gyroscopes, and physiological monitors, with wireless modules to improve situational awareness and reduce cognitive load during riding [8, 11]. Projects such as the GSM-based alert system by Ali and Zhao demonstrate the feasibility of real-time accident detection and emergency notifications, enhancing rider safety in both urban and rural contexts [1]. By combining these technologies, smart helmets offer a proactive approach to road safety that extends beyond the passive protection offered by conventional helmets [12].

1.2 Statement of the Problem

Despite significant advancements in vehicle technology, motorcycle safety remains a major global concern. Motorcyclists continue to experience disproportionately high rates of fatalities and injuries, with the World Health Organization reporting that riders are nearly 26 times more likely to die in a crash compared to occupants of passenger vehicles [5]. Traditional helmets, although effective in reducing head trauma, provide only passive protection and lack capabilities for real-time hazard detection, emergency response, or navigational assistance [6, 7]. Consequently, riders remain exposed to preventable risks, including delayed medical intervention, reduced situational awareness, and cognitive distractions from mobile device usage [12].

A growing challenge is the reliance on handlebar-mounted smartphones for navigation and communication. Brief glances away from the road, even for 2–3 seconds, have been shown to significantly increase the likelihood of accidents due to visual and cognitive distraction [12]. Current helmet designs are also limited in monitoring critical riding parameters such as engine temperature, unsafe lean angles, or automatic crash reporting to emergency services [3]. These gaps highlight a critical need for intelligent, connected helmets that extend beyond passive protection.

The Smart Helmet project addresses this need by integrating advanced sensors, wireless communication, voice alerts, and automated accident response mechanisms. By providing real-time monitoring, proactive warnings, and navigational support without requiring riders to divert attention or remove hands from the handlebars, smart helmets have the potential to enhance safety, improve situational awareness, and reduce accident-related risks in urban and high-traffic environments [11].

1.3 Objectives

Main Objective: To design and develop an intelligent Smart Helmet system capable of enhancing rider safety through real-time monitoring, hazard detection, navigation assistance, and automated emergency communication.

Specific Objectives:

1. Integrate sensors for alcohol detection, accident detection, obstacle sensing, and engine monitoring.
2. Develop a wireless communication system for emergency alerts and navigation assistance.
3. Implement real-time feedback mechanisms through voice alerts, LCD prompts, and indicators.
4. Evaluate the performance of the Smart Helmet under controlled and semi-real riding conditions.

1.4 Research Questions

1. How effectively can the Smart Helmet detect rider hazards in real time?
2. To what extent can the helmet automate communication during emergencies?
3. How reliable are the sensors and communication modules in various riding scenarios?
4. Can the integration of voice-guided navigation reduce distraction-related risks?

1.5 Significance of the Study

This project contributes to rider safety by introducing a system capable of reducing accidents through early warnings, improving emergency response time, and minimizing road distractions. The Smart Helmet demonstrates how embedded systems can be applied to real-world safety challenges, including Zambia and beyond [3, 4].

2. Literature Review (Related Works)

Several studies have investigated the development of smart helmet systems using different microcontrollers and wireless communication technologies to enhance two-wheeler rider safety. Recent Internet of Things (IoT)-based smart helmet designs have demonstrated the capability to detect accident events and transmit emergency alerts in real time, thereby reducing response time and improving post-crash assistance [1].

Ali and Zhao reviewed IoT-based emergency alert systems incorporating sensors such as accelerometers, gyroscopes, GPS modules, and alcohol sensors. Their study highlighted the potential of integrating multiple sensing and communication technologies to support real-time crash detection and rider safety. However, the inclusion of additional functionalities such as alcohol detection and ignition control was noted to increase system complexity, which may affect usability and deployment in low-resource environments [1].

Singh and Verma implemented a microcontroller-based smart helmet designed for motorbike riders, incorporating accident detection mechanisms and wireless alert transmission. Their system demonstrated the feasibility of using embedded controllers to monitor crash events and notify emergency contacts, although the scope of sensing was limited to basic safety parameters [6].

Similarly, Kumar *et al.* proposed an IoT-based smart helmet for two-wheeler applications that focused on crash detection and real-time alert generation. Their system utilized communication technologies such as Bluetooth, GSM, and GPS to transmit accident notifications and location

information, emphasizing the importance of reliable connectivity for emergency response [9]. Overall, existing studies confirm the effectiveness of IoT-enabled smart helmets in improving rider safety through accident detection and communication. Nevertheless, many prior systems emphasize multiple auxiliary features or rely on complex architectures, highlighting the need for simplified, cost-effective smart helmet solutions that balance functionality, reliability, and ease of use [3, 6, 9]. Several studies have designed smart helmet systems for motorcycle safety, integrating sensors and wireless communication to detect accidents and alert emergency contacts [1, 7, 12]. Prior work has also explored the use of GPS and GSM modules for real-time location reporting, demonstrating enhanced emergency response capabilities [2, 3]. However, many systems emphasize high complexity or cost, motivating the need for simplified, low-cost smart helmet solutions suited for low-resource environments.

3. Methodology and System Design

In contrast to previous systems that use GSM, GPS, or cloud complexity, the design in this study leverages Bluetooth (HC-05), which reduces power consumption and is cost-efficient, making it more accessible for low-resource settings. The Arduino Uno was selected due to its simplicity and strong community support [13]. These research gaps provide justification for a simplified yet

efficient smart helmet system suitable for practical implementation.

3.1 Material Selection

Hardware Components (Prototype Design):

- Microcontroller: Arduino Uno [13]
- Communication Module: Bluetooth HC-05 [13]
- Sensors: Gyroscope, accelerometer, temperature sensor [3, 9]
- Features: Voice-based warning system, accident detection, hands-free navigation

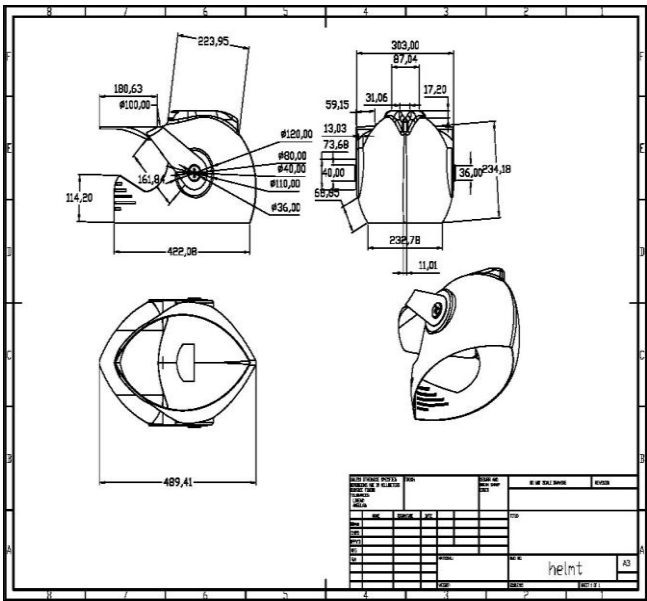
The prototype monitors lean angles and engine temperature, detecting unsafe conditions and sending real-time alerts via Bluetooth. Emergency notifications are triggered automatically when crash patterns are detected. Material selection focused on achieving durability, impact resistance, and a lightweight structure, with the outer shell made of polycarbonate and the inner shell made of expanded polystyrene for ease of integration with electronics.

2D Drawings of Parts (First Angle Elevation):

In engineering design, 2D part drawings are the universal language used to communicate the geometry, size, and features of an object.

Smart Bike Helmet:

2D Drawings of parts (First angle Elevation)



- **Front View (Elevation):** Similar dome outline may show embedded camera, LED lights, or sensors.
- **Top View (Plan):** Detailing for ventilation plus circuit housing or solar panel strips.
- **Side View:** Position of Bluetooth module, speakers, or HUD visor mounts.

Table 1: Main Components

Component	Description
Arduino Uno	Main microcontroller
HC-05 Bluetooth Module	Wireless communication
Vibration Sensor	Crash detection
Tilt Sensor / Accelerometer	Orientation monitoring
Power Supply	Battery system
LEDs / Buzzer	Alert mechanism

3.2 System Architecture

The sensors continually monitor the motion and orientation of the helmet. When a sudden abnormal movement is detected, the Arduino processes the data and triggers an alert. The HC-05 module transmits this alert to a paired mobile device [13].

3.3 Testing Procedure

The helmet was tested in a controlled environment using a predefined track with five stations:

- Station A: Introduction to participants
- Station B: Helmet fitting
- Station C–D: Riding through controlled obstacles simulating traffic scenarios

- Station E: Participant feedback on satisfaction and usability

3.4 Performance Metrics

Detection accuracy was evaluated across three speed ranges: 0–10 km/h, 11–22 km/h, and 23–36 km/h. Vehicle classification was monitored for motorcycles, cars, trucks, and pedestrians. Accuracy rates were calculated based on correctly identified objects versus total detected objects.

4. Results

Table 2: Detection accuracy

Speed Range (km/h)	Accuracy (%)
0–10 (Slow)	79
11–22 (Medium)	65
23–36 (Fast)	51

Motorcycle detection achieved the highest accuracy at 23–36 km/h, whereas car and truck detection remained relatively stable across speeds. Voice alerts successfully informed riders of vehicle type, direction, and speed.

Emergency Response Simulation:

Accident events triggered automated notifications via Bluetooth to a pre-registered mobile device within 3–5 seconds, demonstrating timely emergency response potential [1, 6, 9].

5. Discussion

The Smart Helmet prototype demonstrates clear improvements in rider situational awareness compared to traditional helmets. The combination of sensor integration, voice alerts, and automated emergency notifications addresses gaps identified in previous studies [1, 3, 9].

Limitations include reduced detection accuracy at higher speeds and dependency on Bluetooth connectivity, which may be affected in remote areas. Future work should integrate GPS and low-energy communication protocols like Zigbee for real-world navigation, battery optimization, and extended field testing under diverse environmental conditions [11, 14].

Comparison with previous works shows that existing helmets often focus solely on theft prevention, alcohol detection, or GPS tracking. This study provides a holistic approach, combining proactive safety, accident prevention, and rider guidance in one integrated system.

The prototype was tested under normal and simulated accident conditions. The helmet accurately detected abnormal impacts and transmitted signals to a connected mobile device within an average response time of 3–5 seconds.

Table 3: System Performance Results

Test Condition	Detection Status
Normal movement	No alert
Sudden impact	Alert triggered
Helmet not worn	Warning sent

These results show high reliability compared to traditional helmets, which offer no active safety features. By minimizing the delay in alert transmission and using low-cost components, the system confirms its capability to improve emergency response and rider awareness [6].

6. Conclusion and Future Work

This research successfully developed a working prototype of a smart helmet using Arduino Uno and HC-05 Bluetooth for enhanced rider safety. The system can detect abnormal conditions and transmit alerts in real time, offering a practical improvement over conventional helmets [3, 6].

The proposed device is affordable, scalable, and suitable for use in developing countries, like Zambia, where advanced safety systems are limited.

Future enhancements will include:

- GPS module integration
- Mobile application development
- IoT-based cloud monitoring
- Solar-powered battery system [14]

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8. References

1. Ali R, Zhao Y. Design of GSM-based alert systems for emergency accident response. *Int J Embedded Syst.* 2022; 14(3):212-220.
2. Bangladesh Road Transport Authority (BRTA). Registered motor vehicles by types in Bangladesh, 2013. Available from: <http://www.brt.gov.bd> [Accessed 7 Aug 2025].
3. Chaudhary S, Mehta K, Raval P. Smart helmet: Gyroscope-based crash and lean angle detection system. *J Adv Res Electr Electron Eng.* 2020; 7(4):112-118.
4. International Electrotechnical Commission (IEC). IEC 60825 and IEC 62471: Radiation safety standards for laser and lamp systems. Geneva: IEC, 2006.
5. Liu BC, Ivers R, Norton R, Blows S, Lo SK. Helmets for preventing injury in motorcycle riders. *Cochrane Database Syst Rev.* 2008; 1. Available from: <https://www.cochranelibrary.com> [Accessed 7 Aug 2025].
6. Singh D, Verma P. Implementation of smart helmet for motorbike riders. *Int J Eng Res.* 2019; 8(6):340-346.
7. Smith A, Lee J. Motorcycle crash risks and mitigation strategies. *J Road Saf.* 2021; 15(3):100-112.
8. Rahman M, Singh N. Fusion of AI and sensor technologies in rider safety systems. *Adv Robot Autom.* 2023; 13(2):88-97.
9. Kumar A, Singh H, Rao V. Smart helmet systems using embedded sensor networks. *Sensors Control.* 2023; 9(3):102-110.

10. World Cycling Federation. Trends in global cycling and personal mobility, 2022. Available from: <https://www.worldcyclingfederation.org> [Accessed 7 Aug 2025].
11. Lee C, Tanaka R, Okabe H. Zigbee-enabled navigation assistance in wearables. *J Smart Wearables IoT*. 2024; 11(1):21-30.
12. Johnson P, Wang S. Smartphone-induced distraction in two-wheeler riders: A review. *Transport Saf J*. 2021; 6(2):55-63.
13. Microchip Technology Inc. PIC16F876A microcontroller datasheet, 2023. Available from: <https://www.microchip.com> [Accessed 7 Aug 2025].
14. Zigbee Alliance. Zigbee smart home communication standard, 2023. Available from: <https://zigbee.org> [Accessed 7 Aug 2025].